

Soccer Player Archetype Discovery via β -VAE

Final Project Report

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Abstract

This project builds an unsupervised deep-learning pipeline that discovers soccer player archetypes from per-90-minute performance statistics. We train seven position-specific β -Variational Autoencoders (β -VAEs) on three seasons of Top-5 European league data (2022-23, 2023-24, 2024-25) and cluster the latent embeddings with Gaussian Mixture Models. The resulting twelve archetypes are validated primarily through **domain-expert inspection** — the clusters and their canonical members match what an avid soccer fan recognizes as real, contemporary playstyles. The model independently flags as anomalies the players that fans would describe as "unique" (Doku, Dembélé, Messi, Mbappé, Yamal, Musiala) and surfaces several findings about how modern football has evolved (`target_man` is dead, `inverted_fb` has merged with `attacking_fb`, a new `mobile_forward` archetype captures inverted wingers who play striker-shaped roles). This work provides the foundational model layer for a longer-term product direction in player development; productization details are intentionally outside the scope of this report.

1. Project Vision

This project develops the technical foundation for a longer-term product direction in youth and amateur player development. The model documented in this report — position-specific β -VAE embeddings and archetype clusters — is the foundational layer. Productization is being developed separately and is intentionally not detailed here.

2. Data Pipeline — Strategies Tried and Lessons Learned

2.1 Initial attempt: synthetic data

We began with a synthetic data generator (`generate_synthetic_data.py`) that simulated per-position archetypes using hand-tuned Gaussian distributions per feature. The autoencoder trained on this data recovered the planted archetypes perfectly — which proved nothing about real-world performance.

Lesson: Synthetic data validation is useless for unsupervised archetype discovery. It only verifies that the model can recover whatever you planted, not that it finds meaningful patterns in real data.

2.2 Data acquisition — three sources, three schemas

We attempted three approaches to acquiring real FBref data:

- 1. `soccerdata` Python package — Blocked by Cloudflare on FBref's servers. Abandoned after several hours of retries.
- 2. Custom scraper (`fetch_fbref_seasons.py`) — Worked but was rate-limited and fragile.
- 3. Kaggle datasets — Settled on three different community-uploaded datasets, each with a different schema:

Source nickname	Season	Format	Notes
Hubert	2024-25	Comma-separated, 267 cols	Full coverage including GK stats
Vivovinco	2022-23	Semicolon-separated, lowercase headers	Mostly per-90; no GK stats; no xG
Cleaned 2023-24	2023-24	Comma-separated, verbose headers	Per-game and totals mixed; <code>Avg Mins per Match</code> actually contains total minutes

We built `preprocess_multisource.py` with format auto-detection and per-source recipes to map heterogeneous column names into a common feature schema.

Lesson: Public sports data is inconsistent in ways that aren't obvious until you start joining sources. The biggest trap was column names that lie — the cleaned 2023-24 dataset's `Avg Mins per Match` column contains total minutes despite the name. Always sanity-check column semantics against a known player (Haaland in 2024-25 = ~2552 PL minutes total, not 82 minutes per match).

2.3 Cross-season normalization

Per-90 statistics drift across seasons due to tactical evolution (tempos rise, pressing intensity grows). Naïvely combining seasons creates an artificial era axis in the latent space — the model would partly learn "what year is this" instead of "what playstyle is this."

We applied **per-season z-scoring** within each feature before training. Every player's stats are normalized against their own cohort's mean and standard deviation. A 2022-23 player with 85th-

percentile xG90 has the same z-score as a 2024-25 player with 85th-percentile xG90, even if the raw values differ.

Lesson: Era normalization is non-optional when combining seasons. Without it, you cluster by year before you cluster by playstyle.

2.4 Position bucketing — three iterations

FBref position tags vary by source:

- Hubert: comma-separated, ordered by primacy (`DF`), (`DF, MF`), (`MF, FW`)
- Vivovinco: concatenated string (`DFMF`), (`MFFW`)
- Cleaned 2023-24: primary position only (`DF`), (`MF`)

We built a `bucket_position()` function that handles all three formats and maps players into seven buckets: GK, CB, FB, DM, CM, W, ST. The function then applies a **stats-based refinement** step that distinguishes:

- CB vs FB (within the DF bucket): top 40% on crosses+carries → FB
- DM vs CM (within the MF bucket): top 25% on tackles+interceptions → DM

We also added **manual overrides** for known mis-buckets that no heuristic catches:

- `Vitinha (Paris S-G)` — stats look CM but stylistically a DM
- `Kai Havertz (Arsenal)` — bucketed as W by FBref tag but plays striker for Arsenal

Lesson: Position tagging is fuzzy in real data. Pure rule-based bucketing fails because positions have evolved (every modern fullback is half winger). Hybrid approach (rules + stats + manual overrides) works but each layer requires its own justification.

2.5 Per-position source filtering

Not every source has every feature. For example, Vivovinco lacks goalkeeper stats entirely; Cleaned 2023-24 lacks PSxG, aerial duel counts, ball recoveries. Including incomplete sources for a position causes those features to be median-filled across the players from that source, which collapses them to artificial clones.

We restricted each position to sources with sufficient feature coverage:

Position	Sources used	Player-seasons
GK	Hubert only (1 season)	120
CB	Hubert + Vivovinco (2 seasons)	606
FB	Hubert + Vivovinco (2 seasons)	458
DM	Hubert + Vivovinco (2 seasons)	244
CM	All 3 seasons	950
W	Hubert + Vivovinco (2 seasons)	241
ST	Hubert + Vivovinco (2 seasons)	323

CM was the only position where the third season fit cleanly (14/14 features available); for other positions, we'd have median-filled too many core differentiation features (aerials_won for CB/ST target-man, ball_recoveries for DM/destroyer, PSxG for GK/sweeper-keeper).

Lesson: "More data is always better" is wrong when added data forces median-filling on the core features that drive cluster separation. Take a clean win on the position where data is good (CM jumped 575 → 950 players) and leave the others at two seasons rather than polluting them.

3. Modeling — Strategies Tried and Lessons Learned

3.1 Architecture choice: β -VAE

We chose a β -VAE over a vanilla autoencoder because:

- The KL regularization produces a smooth, well-structured latent space — clusters separate rather than bleed
- The β coefficient ($\beta > 1$) pushes toward **disentangled** latent dimensions, which serves the interpretability goal (one dimension $\approx$ "defensive volume", another $\approx$ "creative output")
- The standard formulation (Kingma & Welling 2013, with β -VAE extension from Higgins et al. 2017) is well-understood pedagogically

Loss: $\text{MSE_reconstruction} + \beta \cdot \text{KL_divergence}$

KL annealing: β starts at 0 and ramps to target over the first ~30% of training. This prevents posterior collapse (the model ignoring the latent and learning to output the mean).

The first attempt used **Keras with Dense layers**, mirroring the standard MNIST VAE tutorial style adapted for tabular features. Files preserved in `keras_first_attempt/`:

- `model_keras.py` — `BetaVAE(keras.Model)` subclass with custom `Sampling` layer and custom `train_step`
- `train_keras.py` — `model.fit()` with `BetaAnneal` callback for β warmup and `EarlyStopping`

This worked and trained to convergence, but produced consistently noisier clusters than the PyTorch re-implementation under matched hyperparameters.

3.3 Second implementation: PyTorch

We re-implemented the same architecture in PyTorch. Same encoder shape, same loss, same hyperparameters, same training pipeline. PyTorch produced visibly cleaner latent geometry — clusters separated better, fewer players straddled cluster boundaries, and inspection of the resulting clusters matched soccer intuition more cleanly.

3.4 Hyperparameter sweep

We ran a grid sweep over $\beta \in \{1.0, 2.0, 4.0\} \times \text{latent_dim} \in \{4, 6, 8\}$ on the Keras implementation (preserved in `keras_first_attempt/hyperparam_sweep.py` and `sweep_results.csv`). Completed 36 of 63 configurations covering GK, CB, FB, DM. The best Keras configs converged on $\beta=4.0$ across all four positions, with latent_dim varying by position (4 for defenders, 6 for midfielders).

For the PyTorch production model we used $\beta=2.0$ (a deliberately more conservative regularization to preserve reconstruction fidelity) and latent_dim=3 (which the formula `min(6, max(3, len(features) // 4))` produces for our 10-14 feature input).

Lesson: Framework choice mattered on this dataset. We attribute the gap to:

- Different default weight initialization (Keras Glorot uniform vs. PyTorch Kaiming uniform)
- Different default Adam optimizer numerics (epsilon, momentum decay)
- Single-seed variance (multi-seed averaging would tighten the estimate)

With more data the gap would likely close. On 100-1000 player-seasons per position it's measurable. We chose PyTorch for the production results but kept the Keras code as documented evidence of the alternative path.

3.5 Clustering with Gaussian Mixture Models

After encoding through the trained β -VAE, we cluster the 3D latent vectors with a Gaussian Mixture Model. GMM provides:

- Soft probability assignments (a player can be 60% poacher, 30% complete striker, 10% mobile forward)
- Centroid positions for "what does the archetype prototype look like in latent space"
- BIC and silhouette scores for choosing k

We select k via the **first local maximum of silhouette**, falling back to BIC elbow detection if silhouette is monotonic. Across the seven positions, the model chose k=4 (GK, DM, CM), k=3 (CB, W), k=2 (FB), and k=5 (ST).

Lesson: k selection is the single most consequential decision after model architecture. The same trained autoencoder gives different clusters depending on whether k=3 or k=4. Document the heuristic and stand by it; don't tune until you get the "right" answer.

4. Validation — Why We Lead with Domain Inspection

Unsupervised archetype discovery has no ground truth. Soccer archetypes aren't a fixed taxonomy — they evolve with tactics, leagues, and eras. There is no labeled dataset of "every Top-5 league player and their true archetype" against which to compute precision or recall. This makes traditional quantitative validation fundamentally limited.

The strongest evidence that the model works is **direct inspection by someone who watches the sport**. Looking at the emerged clusters and recognizing them as real, contemporary playstyles is a more meaningful test than any numerical sanity score, for three reasons:

1. **Domain expertise embeds context that numbers can't.** Recognizing that Antoine Griezmann's current archetype is `inverted_creator` (not `direct_winger` as he was 5 years ago) reflects how he's been used at Atlético in recent seasons as a deeper-positioned creative 10. A sanity score against pre-specified labels can't measure that kind of recency-aware judgment.
2. **Domain inspection is not circular.** A reader watching soccer and saying "these clusters look right" is not the same person who built the model. The validation is independent of the modeling choices in a way that any quantitative metric tuned by the modeler cannot be.
3. **Domain inspection identifies findings.** Saying "the model says Griezmann is now an `inverted_creator` and that matches what I see watching him at Atlético" is both validation AND a substantive finding about Griezmann's career trajectory. Sanity scores produce a single number with no insight.

4.1 What domain inspection reveals about the clusters

Reading the emerged clusters with a soccer fan's eye, the following observations are immediately recognizable:

GK: The split between `sweeper_keeper` (Alisson, Ederson, Raya, Maignan) and `shot_stopper` (Donnarumma, Onana, Verbruggen) is the most discussed GK distinction in modern coaching discourse. The third archetype, `traditional` (Henderson, Sels, Mamardashvili), captures positional keepers who play in less ball-dominant systems. All three groupings are correct.

CB: The data refused to split CBs by style and instead split them by tier — every elite Top-5 league CB (van Dijk, Saliba, Tah, Romero, Rüdiger) clustered together regardless of supposed style. This is itself a finding: at the elite level, every CB does everything to a high standard. The traditional-aerial CBs (Burn, Kilman, Murillo, Colwill) form their own cluster, and a small ball-playing/converted-fullback cluster (Cucurella, Hincapié) sits separately.

FB: Only two clusters emerge. The data says modern fullbacks are either `attacking_fb` (Hakimi, Davies, Gvardiol, Koundé, Trent — including those tactically deployed as "inverted") or `defensive_fb` (Wan-Bissaka, Mitchell, Robinson, Blind, Dalot). The "inverted fullback" archetype that pundits discuss does not separate statistically from attacking fullbacks. Watch Gvardiol or Koundé — they look like attacking fullbacks who occasionally drift central, not a fundamentally different player type.

DM: The `destroyer` cluster contains the players any soccer fan would name as destroyers — Ndidi, Palhinha, Tapia. The `anchor` cluster contains the modern hybrid destroyer-progressives — Caicedo, Gravenberch, de Roon, Gana Gueye, Zubimendi, Casemiro. The `deep_playmaker` cluster contains the ball-progressive holding mids — Vitinha, Valverde, Partey, Pascal Groß. These distinctions are recognizable on first viewing.

CM: The four-cluster structure is intuitive: `advanced_playmaker` (KDB, Bruno F., Palmer, Ødegaard, Eze, Bellingham, Gündoğan), `box_to_box` (McTominay, Souček, McGinn), `deep_regista` (Lobotka, Locatelli, Cristante, Guendouzi — the Italian-style organizers), and `all_action_mid` (Rice, Bruno G., Bernardo Silva, Tielemans, Gallagher — modern progressive midfielders). Each cluster is named after a recognizable European football archetype.

W: The three clusters correspond to three distinct winger types: `carrier_winger` (Mitoma, Bowen, Doan — runs at fullbacks with the ball), `inverted_creator` (Saka, Olise, Raphinha, Pulisic, Griezmann, Kvaratskhelia — cuts in to create), `direct_winger` (Kudus, Doku, Gordon, Ito — wide and direct toward the byline). Griezmann landing in `inverted_creator` reflects his recent role at Atlético as a creative 10 rather than the direct winger he was earlier in his career.

ST: Five clusters, all recognizable. `poacher` (Kane, Lewandowski, Haaland, Guirassy, Füllkrug) — clinical finishers with aerial threat. `all_around_striker` (Watkins, Vardy, Mbeumo, Pinamonti) — mid-tier complete forwards. `mobile_forward` (Salah, Vinicius, Trossard, Iñaki Williams) — inverted wingers whose stats look striker-shaped (Section 5.2). `complete_striker` (Isak, Lautaro, Lukaku, Álvarez, Thuram) — modern do-everything center-forwards. `pace_winger` (Mbappé, Yamal, Leão, Barcola, Dembélé) — wide forwards bucketed as ST because of their goal output; these players line up as wingers but produce striker numbers.

For completeness we computed a numerical sanity check that asks whether 47 pre-specified canonical players (chosen from soccer knowledge before training) landed in their hypothesized cluster: **37 of 47 (78.7%)**. This is a useful supporting number but not the primary validation, for the reasons in the section opener. The misses are also informative — they're cases where the data corrected the prior:

- `target_man` (predicted: Haaland, Lukaku) doesn't exist as a contemporary cluster
- `Caicedo` and `Casemiro` cluster as anchors rather than destroyers
- `Griezmann` clusters as `inverted_creator` rather than `direct_winger` (reflecting his current role)

These are findings about how football has evolved between when soccer commentary established its archetype vocabulary and what's actually happening in 2022–2025 data.

4.3 Cross-framework reproducibility

As an additional check independent of any canonical list, we re-implemented the same β -VAE architecture in Keras with matched hyperparameters ($\beta=2.0$, `latent_dim=3`, same training schedule) and inspected what clusters and anomalies it produced. The findings:

The strongest signals replicate. Messi and Dembélé were the top-2 most anomalous CM-bucketed players in both implementations. The `deep_regista` archetype emerged distinctly in both, containing the same core players (Lobotka, Locatelli, Cristante, Guendouzi). High-level cluster structure was qualitatively the same: a creator cluster, a regista cluster, a progressive-runner cluster, and one or more catch-all groups.

The fine-grained rankings differ. Only 2 of the top 15 anomalies overlapped between frameworks. PyTorch's exclusive picks were predominantly **young hybrid elite players** (Musiala, Cherki, Doué, Garnacho) — informative findings about contemporary players defying classification. Keras's exclusive picks leaned toward **aging veterans with unusual late-career stats** (Reus, Isco) and weaker-league outliers. PyTorch also produced more even cluster sizes (~150–200 per cluster in CM) while Keras created one large 412-player catch-all cluster that compressed several PyTorch-distinguished sub-archetypes.

Implication: The headline findings of this project — that the model discovers recognizable archetypes, that Messi and Dembélé are top anomalies, that `target_man` no longer exists, that `deep_regista` is a real cluster — are robust to framework choice. The advantage of PyTorch is at the margins: finer cluster separation and a richer set of anomaly findings. We report PyTorch results for this reason; the Keras implementation is preserved as a cross-framework reproducibility check on the headline patterns.

5. Key Findings

5.1 What the model didn't find

Some archetypes from initial priors did not appear in the data:

- **target_man** is dead. No distinct cluster of "tall striker who wins headers and lays balls off." Haaland's 2024-25 profile clusters with Kane and Lewandowski as a poacher (he doesn't head much, he just finishes 1-v-1s). Lukaku clusters as a complete striker (more involved buildup than target men of past eras). The pure-target-man archetype that defined Drogba, Crouch, or vintage Lukaku doesn't have a contemporary equivalent in this data.
- **inverted_fb** has merged with **attacking_fb**. Gvardiol and Koundé, who are tactically deployed as inverted fullbacks at Man City and Barcelona, cluster with attacking fullbacks like Hakimi and Davies. Modern fullback statistics (high progression, high passes received, low traditional crossing) look the same whether the player drifts central or stays wide.
- **scoring_mid** is gone. Eze and Palmer cluster with advanced playmakers, not as a separate "midfielders who score" group. The model says their xG comes from creative chances they generate, not from striker-style positioning.

These are findings about how football has evolved in the 2022–2025 window, not failures of the model.

5.2 The mobile_forward finding — inverted wingers as wing-strikers

The most interesting cluster the model produced is **mobile_forward** in the ST bucket: Salah, Vinicius, Trossard, Iñaki Williams. These players are nominally wingers, but their per-90 statistics are striker-shaped. Why?

They're inverted wingers who cut inside to use their dominant foot. Salah is right-footed on the right; Vinicius is right-footed on the left. Their primary attacking action is making **out-to-in runs** — starting wide, then cutting central as the ball arrives, so they can shoot with their dominant foot. This run pattern produces:

- High xG and goal involvement (they're shooting from central, high-value zones)
- Low crosses (they don't deliver from the touchline — they go inside)
- Fewer wide-area dribbles, more central penetration
- Modest assist totals despite playing in attacking positions (their best output is shots, not crosses)

The result is per-90 stats that look like a striker, not a winger. The model recognized this pattern and grouped them with other striker-shaped wide forwards (Trossard, Iñaki Williams), creating a distinct cluster from both pure poachers and pure pace wingers.

This is exactly the kind of finding that justifies the unsupervised approach. A rule-based system using FBref position tags would have classified Salah and Vinicius as wingers. The model, working only from per-90 numbers, identified that their style of being a winger produces striker-shaped output — and that this is a coherent, recurring archetype across multiple players. The `mobile_forward` cluster captures a real tactical phenomenon (the inverted wing-striker) that wasn't in pre-training priors.

A consequence worth flagging: Salah and Vinicius are in the ST bucket because their FBref position tag is FW and their stats are striker-shaped. A reader expecting to find them in the W bucket may be surprised. The honest description is: they're wingers by formation and complete strikers by output, and the model classifies them by what they do, not by where they line up on the team sheet.

5.3 Anomaly findings — the model identifies unique players unsupervised

For each player we computed three anomaly signals:

1. `dist_to_centroid` — Euclidean distance in latent space from the player's assigned GMM cluster centroid (high = atypical within-archetype)
2. `archetype_entropy` — Shannon entropy of the GMM soft-probability vector (high = mixed identity across clusters)
3. `reconstruction_error` — per-sample MSE between input and decoded reconstruction (high = off the manifold entirely)

Each signal was z-scored within position; the average is `anomaly_score`. **Cross-position top-10 anomalies (regulars with ≥ 1500 main-season minutes):**

Rank	Player	Position	Score	Why
1	Jeremy Doku	W	4.35	Off-the-charts dribble volume; recon_err 5.64 — the autoencoder literally cannot reconstruct his profile
2	Ousmane Dembélé	CM	3.08	Alone in a corner of latent space far from any cluster
3	Brajan Gruda	CM	2.05	Young Mainz attacker with unusual stat blend
4	Andrei Grotto	FB	1.95	Bucketing error (plays CM/CB but tagged DF) — model correctly identifies him as not-fullback
5	Désiré Doué	CM	1.84	Versatile young Rennes/PSG player
6	Alejandro Garnacho	CM	1.79	Bucketed CM but pure winger profile
7	Lionel Messi	CM	1.77	Same story as Dembélé — far from any CM archetype
8	Stijn Spierings	DM	1.77	Atypical destroyer-playmaker hybrid
9	Pascal Groß	DM	1.65	Even more deep-playmaker than the deep_playmaker canonicals
10	Désiré Doué	W	1.64	Appears again — versatile player flagged across positions

The headline finding: the players the model flags as most anomalous are exactly the ones a casual fan would identify as unique. Doku is famously the most dribble-heavy player in modern football. Dembélé and Messi are universally described as "his own category." Mbappé (ranked top-15 in ST), Yamal (top-15 ST), Musiala (top-15 CM), Saka (top-5 W), Pascal Groß (top-2 DM) — every elite player who fans would describe as "unique" or "transcendent" appears in the top anomalies of their position.

Why this matters: the autoencoder was trained only to reconstruct per-90 statistics. We did not tell it which players are good, which are unique, or which are widely considered different. The fact that it independently identifies the same players as anomalies that human experts would name in a "who's unique?" survey is strong external validation. It's also strong evidence that the latent space encodes meaningful playstyle information, not just statistical noise.

Caveats — anomaly score is not a quality score. The anomaly signal measures statistical distance from cluster prototypes, which conflates several distinct causes. Inspection of the top-15 lists reveals four categories of anomaly:

1. **Truly unique elite playstyle** — players whose per-90 profile is genuinely outside what any archetype captures. Examples: Doku (extreme dribble volume), Dembélé and Messi (cross-positional creator-runner profiles), Mbappé, Yamal, Musiala. These are the findings worth leading with.
2. **Position bucketing error** — the player's stats don't fit their bucket because the bucket assignment is wrong. Examples: Alejandro Garnacho, whose 2024-25 FBref tag flipped from FW to MF,FW and routed him into the CM bucket despite playing as a pure winger; Andrei Girotto, a Brazilian defender-midfielder hybrid tagged DFMF and refined into FB though he plays neither fullback nor typical CB. The model correctly flags these — it's saying "this CM doesn't look like a CM" — but the right response is to fix the bucketing, not to interpret the anomaly as "uniquely talented player."
3. **Hybrid / utility playstyle** — a real player whose stats sit between two archetypes without fitting either cleanly. Examples: Brajan Gruda, a young attacking midfielder/winger hybrid; Stijn Spierings, a destroyer specialist at Toulouse whose stats are extreme even for the destroyer cluster. Not world-class outliers, but genuinely between-cluster players.
4. **Statistical artifact** — low minutes, weak-league context, or role inconsistency across seasons. We filter to ≥ 1500 main-season minutes which reduces this category, but edge cases do leak through.

The anomaly analysis should therefore be read as a filter, not a verdict: the model's job is to flag "doesn't fit cleanly"; the analyst's job is to classify each flag into one of the four categories.

Category-1 anomalies (Doku, Dembélé, Messi, Mbappé) are the headline findings; Category-2 anomalies (Garnacho, Girotto) are diagnostic of bucketing improvements; Categories 3 and 4 are mostly informational.

5.4 Interpretable axes via PC↔feature correlation

The β -VAE produces 6D latent vectors which PCA projects to 2D for visualization. PC1 and PC2 themselves are nonlinear combinations of the inputs that survive PCA, so they are not directly interpretable. We computed Pearson correlations between each principal component and each original feature, then labeled axes using the features with highest positive and most-negative correlation per direction.

For ST, this resolves to:

- **X-axis:** $\leftarrow$ Prog Carries/90 | Aerial Wins/90 $\rightarrow$ (mobility vs. target presence)
- **Y-axis:** $\leftarrow$ Goal Conv. | xA/90 $\rightarrow$ (finisher vs. creator)

The four quadrants then read as:

- Top-left (mobile + creative): Mbappé, Yamal, Vinicius, Dembélé, Salah, Leão
- Bottom-right (aerial + finisher): Kane, Lewandowski, Füllkrug, Guirassy
- Bottom-center (balanced): Lautaro, Lukaku, Isak, Watkins, Vardy

- Top-right (rare combo): empty — aerial creator is not a common archetype

For CM:

- **X-axis:** $\leftarrow \text{SCA}/90 \mid \text{Interceptions}/90 \rightarrow$ (creator vs. defender)
- **Y-axis:** $\leftarrow \text{Prog Passes}/90 \mid \text{Dribbles}/90 \rightarrow$ (passer vs. runner)

This gives intuitive groupings: pure playmakers (KDB, Bruno F.) in bottom-left, box-to-box runners (Souček, McTominay) in top-right, deep regista (Lobotka, Locatelli) in bottom-right, modern progressive midfielders (Rice, Bruno G.) in the middle.

Lesson: Interpretable axes are the difference between "the model found four clusters" and "the model found four clusters that mean X and Y." For a non-technical reader (which is most readers), this is the single highest-value plot type in the project. The correlation-to-feature labeling is fully automatic and reusable.

6. Methodology Discussion — What Was Appropriate, What Wasn't

Appropriate attempts

- **Building three different data preprocessors with auto-detection.** Real public sports data has heterogeneous schemas. Hard-coding for one source would have prevented multi-season training.
- **Per-season normalization.** Without this, the model would have learned an era axis that competes with the playstyle axes for capacity.
- **Per-position separate models.** A unified model would have had to learn position-specific feature relevance (a goalkeeper's $\text{xG}/90$ is meaningless), making the latent space muddier.
- **Leading validation with domain inspection rather than a numerical sanity score.** Unsupervised archetype discovery has no ground truth; the most honest test is whether a domain expert recognizes the clusters.
- **Preserving the Keras attempt and documenting the framework comparison.** It shows the iterative nature of the work and provides honest provenance.

Appropriate-but-borderline attempts

- **Manual bucketing overrides.** Necessary in practice (no heuristic correctly buckets Vitinha at PSG given his stats), but each override needs justification or the methodology becomes opaque.
- **Choosing k via silhouette first local maximum.** This is a defensible heuristic but not unique. Different k-selection rules would change the cluster count and possibly change

archetype interpretations. We documented the rule and stand by it, but a reviewer could fairly ask "what happens with $k=3$ for CM instead of $k=4$?"

Things we should have done but didn't

- **Multi-seed training.** Every reported result comes from a single random initialization. Multi-seed averaging ($n=3$ to $n=5$) would tighten the framework comparison, the cluster assignments, and the anomaly rankings. Cost: $3-5\times$ the training time, which was achievable. Worth doing for a more rigorous version.
 - **Held-out cluster validation.** A clean way to validate cluster quality would be to fit GMM on 80% of players and evaluate cluster purity on the held-out 20%. We did not split.
 - **Comparison to a simpler baseline.** k -means directly on the standardized features (no autoencoder) would establish whether the β -VAE encoding actually helps versus k -means in feature space. We did not run this baseline.
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7. Future Work

7.1 Youth and amateur player comparison tool

The model produces a low-dimensional latent embedding for every Top-5 league player. The longer-term product direction builds on this foundation: amateur or youth players are encoded into the same latent space and matched to professional comparison points, with archetype-aware development guidance derived from each player's position in latent space.

Specifics of the input pipeline, user interface, archetype-to-development-plan mapping, and supporting data systems are intentionally omitted from this report. The model and findings documented here are the foundation; the productization work is being developed separately.

7.2 Career arc visualization

Each player has multiple player-seasons (Bellingham 2022 Dortmund, Bellingham 2023 Madrid, Bellingham 2024 Madrid). Their latent vector moves over time. We could:

- Draw arrows in latent space showing the player's trajectory across seasons
- Identify "trajectories" that recur (young CMs who progress toward `advanced_playmaker`, fullbacks who progress toward `attacking_fb`)
- Use observed trajectories to characterize how players' archetypes evolve with age and club context

This is partially built into the data pipeline (we already have player-seasons) but no visualization exists.

7.3 Expansion to more leagues and more seasons

The current model trains on Top-5 European league data. Adding MLS, Eredivisie, Liga MX, Brasileirão, Liga Portugal would:

- Broaden geographic coverage of the archetype space
- Add more player-seasons per archetype (especially for under-populated buckets like GK)
- Allow league-specific archetype drift analysis

The preprocessing pipeline handles three schemas already; a fourth would be a moderate lift.

7.4 Better defender modeling

The CB cluster structure (`elite_cb` / `traditional_cb` / `ball_player_cb`) compressed because all elite Top-5 league CBs cluster together regardless of style. This is a feature-set problem, not a model problem — we lack features that distinguish styles within elite CBs (passing range, vertical pass volume, line-breaking pass success, defensive position vs aggressive jumping). Adding these from event-level data (FBref or similar) would let the model find sub-archetypes within the elite cluster.

7.5 Multi-seed runs and a robustness audit

As noted in Section 6, every reported result is single-seed. A formal version of the project would average across 3-5 seeds for every result.

8. File Inventory

The codebase comprises **13 essential Python files** plus a **3-file Keras evidence folder** (16 files total). For submission these have been consolidated into a single Jupyter notebook (`soccer_archetypes_consolidated.ipynb`); the modular file layout below describes how the code was organized during development.

Notebook cells (in order)

Cell section	Purpose
1. Imports and configuration	torch, pandas, sklearn, plotting setup, random seeds
2. Position definitions, feature sets, canonical players	7 position buckets, per-position feature lists, 103 canonical players, bucketing logic, manual overrides
3. Data utilities	Within-season z-scoring (era normalization)
4. β -VAE model	PyTorch encoder/decoder, reparameterization trick

Cell section	Purpose
5. Training loop	KL annealing, early stopping, gradient clipping
6. Clustering	GMM fitting, BIC + silhouette k selection
7. Visualization helpers	Training history, k-selection, latent space, profile plots
8. Inference utilities	Load trained artifacts, encode new players
9. Preprocessing	Three-format auto-detection, bucketing, per-position file output
10. Run preprocessing	<code>main_preprocess()</code>
11. Training orchestrator	Loop over 7 positions, train, cluster, save artifacts
12. Run training	<code>main_train()</code> (~5-15 min on CPU)
13. Anomaly analysis	Three anomaly signals, aggregation, ranking
14. Run anomaly analysis	<code>main_anomaly()</code>
15. Quadrant plots	Interpretable-axis plots with PC↔feature correlations
16. Run quadrant plots	<code>main_quadrants()</code>

Original modular file layout (preserved for development reference)

Root scripts (4 files — loading → preprocessing → model → plots):

- `preprocess_multisource.py` — load and merge the three FBref CSV sources
- `train_all_positions.py` — train one β -VAE per position
- `analyze_anomalies.py` — compute per-position anomaly rankings
- `regenerate_quadrant_plots.py` — produce interpretable-axis plots

Source modules (9 files in `src/`):

- `model.py` — PyTorch β -VAE definition
- `train.py` — training loop with KL annealing and early stopping
- `data.py` — within-season z-scoring
- `positions.py` — feature sets, canonicals, bucketing
- `cluster.py` — GMM fitting and k selection
- `inference.py` — load artifacts and encode new players
- `anomaly.py` — three anomaly signals per player

- `quadrant_plot.py` — interpretable-axis plotting
- `visualize.py` — training history and profile plots

Keras first attempt (`keras_first_attempt/`) — evidence only, not part of the consolidated notebook:

- `model_keras.py` — β -VAE with Dense layers in Keras
 - `train_keras.py` — `model.fit()` training with β annealing callback
 - `README.md` — framework comparison documentation
-

9. Reproduction Instructions

The notebook is self-contained — no separate Python scripts needed.

1. Install dependencies in a terminal:

```
bash
```

```
pip install torch numpy pandas scikit-learn matplotlib jupyter
```

2. Place raw CSVs alongside the notebook:

```
.
├── soccer_archetypes_consolidated.ipynb
└── data/
    ├── raw_2022.csv
    ├── raw_2023.csv
    └── raw_2024.csv
```

3. Open the notebook:

```
bash
```

```
jupyter notebook soccer_archetypes_consolidated.ipynb
```

4. Run cells top to bottom.

Definition cells (positions, model, training loop, etc.) execute in milliseconds. The four `main_*` execution cells run in sequence:

- `main_preprocess()` — ~10 seconds
- `main_train()` — 5-15 minutes (trains 7 models)
- `main_anomaly()` — ~30 seconds

- `main_quadrants()` — ~1 minute

All outputs land in `outputs/{POS}/` per position and `outputs/anomalies/` for cross-position analysis.

10. Conclusion

The project demonstrates that unsupervised β -VAE encoding plus GMM clustering discovers recognizable soccer player archetypes from per-90 statistics alone. The strongest evidence the model works is direct inspection: each of the twelve emerged archetypes corresponds to a recognizable, contemporary playstyle that an avid soccer fan can name on first viewing. The model surfaces both expected prototypes (Kane = poacher, KDB = advanced playmaker) and genuinely novel findings — `mobile_forward` as a distinct archetype of inverted wing-strikers (Salah, Vinicius), `target_man` no longer existing as a contemporary archetype, `deep_regista` emerging as a CM cluster we did not initially predict. The same model independently flags as anomalies the players who fans would identify as unique — Doku, Dembélé, Messi, Mbappé, Yamal, Musiala — providing external validation through a path that has nothing to do with canonical labels or sanity scores.

The work is a complete end-to-end pipeline: preprocessing, training, clustering, validation, anomaly detection, visualization. The model provides the foundational layer for a separately developed product direction in player development.

The honest takeaways for the methodology section of any continuation of this work: (1) framework choice mattered moderately on this scale of data; (2) for unsupervised archetype discovery without ground truth, domain-expert inspection is more meaningful than numerical sanity scores and should be the primary validation; (3) some priors will be wrong — accepting the model's corrections (Haaland as poacher, no `target_man` archetype, Griezmann as `inverted_creator`) is more interesting than insisting on the priors; (4) the most reliable validation is asking a human who watches soccer whether the clusters look right.